

Homework 5 – Due by 11:59 PM on Sunday April 27

Instruction: You need to submit your MATLAB (.m) or Python (.py or .txt) files and all the plots. For full credits, the output of your codes should include the running time and the specified plots.

(1) Consider the following logistic regression problem:

$$\min_{x \in \mathbb{R}^m} \frac{1}{n} \sum_{i=1}^n \log(1 + \exp(-b_i a_i^T x)) + \lambda \|x\|_1,$$

where $x \in \mathbb{R}^m$, $a_i \in \mathbb{R}^m$, $b_i \in \{-1, 1\}$ are the data samples associated with a binary classification problem. Fix the seed to 123. Load the `mushrooms.mat` dataset to get matrix $A = [a_i^T]_{i=1}^n \in \mathbb{R}^{n \times m}$ and vector $b = [b_i]_{i=1}^n \in \mathbb{R}^n$, where $n = 8124$ and $m = 112$. Set $\lambda = 10^{-5}$ and $x_0 = \mathbf{0} \in \mathbb{R}^m$. Define $L \triangleq \max_i \{L_i\}$, where L_i is the Lipschitz constant of $\nabla f_i(x)$, and $f_i(x) = \log(1 + \exp(-b_i a_i^T x))$.

(a) Implement Proximal SGD with a single batch size and diminishing stepsize $\alpha_k = 0.1/\sqrt{k}$ for $10n$ iterations. Plot $\|x_k - x_{k-1}\|/\alpha$ versus 10 epochs. Compute and print the running time for running the `for` loop of the algorithm.

(b) Implement Proximal SAGA with stepsize $\alpha = 1/(3L)$. Run the method for $10n$ iterations and plot $\|x_k - x_{k-1}\|/\alpha$ versus 10 epochs. Compute and print the running time for running the `for` loops of the algorithm.

(c) Implement Proximal SVRG++ stepsize $\alpha = 1/(3L)$. Run the method for 10 outer iterations and n inner iterations. Plot $\|x_k - x_{k-1}\|/\alpha$ versus 10 epochs, where x_{k-1} and x_k are the last iterate points of each inner loop. Compute and print the running time for running the `for` loops of the algorithm.

Students enrolled in SIE 549 must solve the following problem.

Students in SIE 449 will get extra credit by solving it.

(2) Consider SVRG++ Algorithm to solve

$$\min F(x) \triangleq f(x) + h(x),$$

where $f(x) = \frac{1}{n} \sum_{i=1}^n f_i(x)$, f has L -Lipschitz continuous gradient, f_i s are convex and differentiable and h is convex but possibly nonsmooth. Show that:

$$\mathbb{E}[\|g - \nabla F(x_t)\|^2] \leq 4L (F(x_t) - F(x^*) + F(\tilde{x}) - F(x^*)).$$

Hint: Use the proof technique from the SVRG algorithm. However, note that in this case, we do not have $\nabla f(x^) = 0$. Instead, suppose $\tilde{h}(x) \in \partial h(x)$. If x^* is an optimal solution, then it must satisfy the optimality condition:*

$$\nabla f(x^*) + \tilde{h}(x^*) = 0.$$

Hint for MATLAB:

```
seed = 123;
rng(seed, 'twister');
load('mushrooms.mat');
[n,m] = size(A);
lam = 1e-5;
% To compute running time:
L = max(vecnorm(A'))^2/4;
tic
...
time1 = toc;
```

Hint for Python:

```
import numpy as np
import scipy.io
import matplotlib.pyplot as plt
import time
import scipy.sparse as sp
np.random.seed(123)
data = scipy.io.loadmat('mushrooms.mat')
A = data['A']
b = data['b']
n, m = A.shape
lam = 1e-5;
L = 0
for i in range(n):
    L = (np.maximum(L, sp.linalg.norm(A[i,:])))
L = L**2/4
# To compute running time:
start_time = time.time()
...
time1 = time.time() - start_time
```